

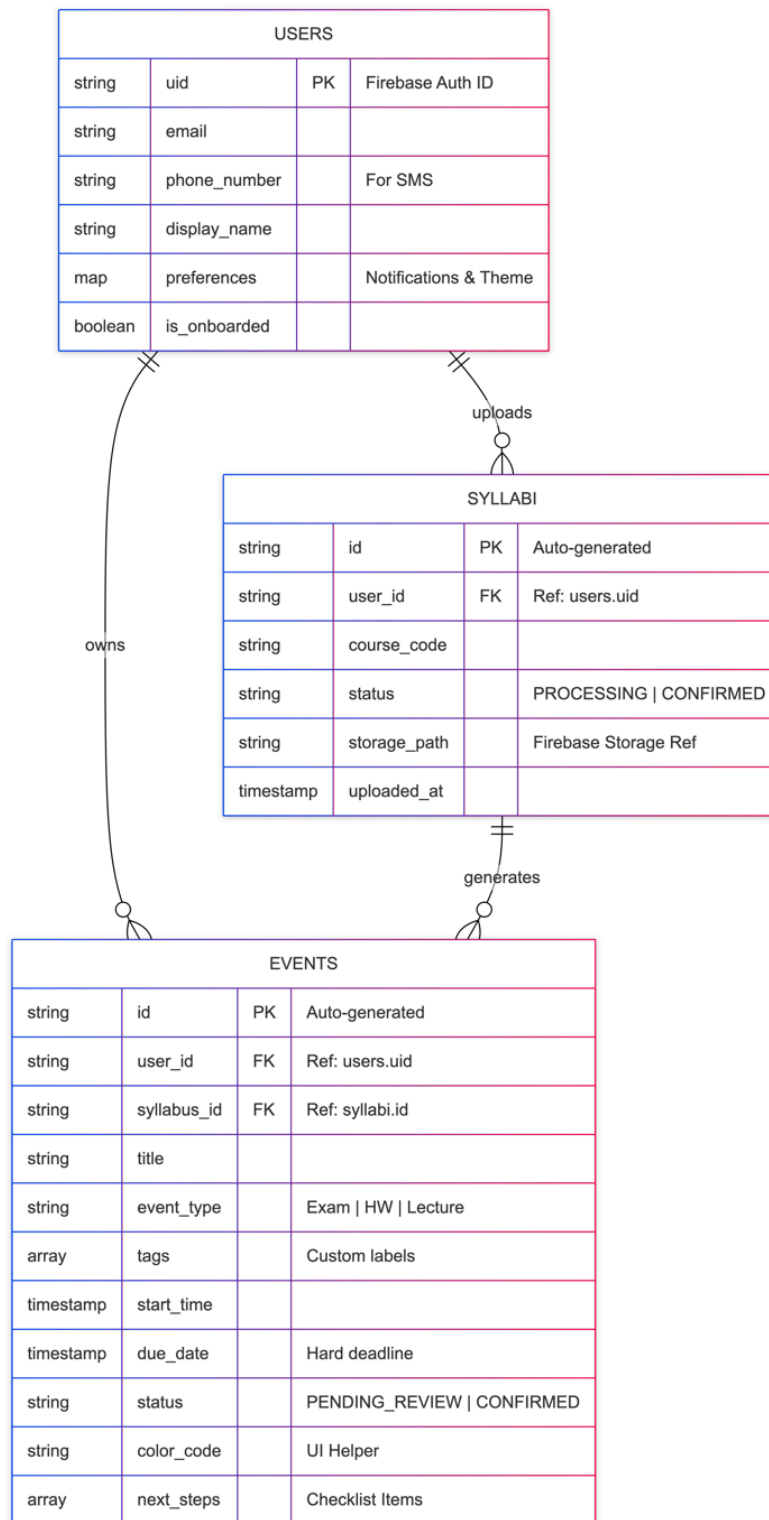
Technical Spec: Database Schema (v1)

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1. Overview

This document defines the Firestore NoSQL schema for the CoursePilot MVP. The structure is designed to support the **Async AI Pipeline** and **Real-time Dashboard** updates.

2. Logical Entity Relationship Diagram



3. Collection Definitions

A. Users Collection

- **Collection ID:** users
- **Purpose:** Stores profile data and global notification settings.
- **Key Design:** Preferences are grouped to simplify Notification Service reads.

```
1 // Collection: users
```

```

2  {
3    "uid": "abc123456789",
4    "email": "user@usc.edu",
5    "phone_number": "+12130000000",
6    "display_name": "John Doe",
7    "photo_url": "https://...",
8    "created_at": "2026-01-20T10:00:00Z",
9
10   "preferences": {
11     "notifications": {
12       "email_enabled": true,
13       "sms_enabled": true, // If true, backend checks phone_number
14       "push_enabled": true,
15       "timing": {
16         "one_week_before": true,
17         "three_days_before": true,
18         "twenty_four_hours_before": true,
19         "same_day": false
20       }
21     },
22     "theme": "light",
23     "calendar_start_day": "Sunday"
24   },
25
26   // ADDED: Tracks if they have finished the initial setup/onboarding
27   "is_onboarded": false
28 }
29

```

B. Syllabi Collection

- **Collection ID:** syllabi
- **Purpose:** Tracks the async processing pipeline.
- **Key Field:** status (The Frontend listens to this field to show the progress bar).

```

1  // Collection: syllabi
2  // Document ID: Auto-generated (e.g. syl_98765)
3  {
4    "id": "syl_98765",
5    "user_id": "abc123456789",
6    "course_code": "CSCI 570",
7    "file_name": "CSCI570_Syllabus.pdf",
8    "storage_path": "users/abc123/syllabi/CSCI570.pdf",
9
10   // Pipeline Status: UPLOADED -> PROCESSING -> PENDING_REVIEW ->
   CONFIRMED
11   "status": "PROCESSING",
12
13   "uploaded_at": "2026-02-05T14:30:00Z",
14   "metadata": {
15     "instructor": "Dr. Shamsian",
16     "semester": "Spring 2026"
17   }
18 }

```

C. Events Collection

- **Collection ID:** events
- **Purpose:** The core data for the Calendar Dashboard.

Key Design: This collection is **self-contained**. It includes `course_code` and `color_code` directly so the frontend can render agenda cards without performing expensive joins.

```

1  // Collection: events
2  {
3    "id": "evt_55566",
4    "user_id": "abc123456789",
5    "syllabus_id": "syl_98765",
6
7    "title": "Assignment 1",

```

```
8  "course_code": "CSCI 570",
9  // ADDED: Supports custom grouping (e.g. "Urgent", "Group Project")
10 "tags": ["CS402", "Urgent"],
11 "color_code": "#44BDD0",
12 "event_type": "HW",
13
14 // TIMING UPDATES
15 // start_time = When the calendar block starts (e.g. "Work on
assignment")
16 "start_time": "2026-02-20T14:00:00Z",
17 "end_time": "2026-02-20T16:00:00Z",
18 // ADDED: The actual hard deadline (often different from the
calendar block)
19 "due_date": "2026-02-20T23:59:00Z",
20 "all_day": false,
21
22 "status": "CONFIRMED",
23 "location": "Brightspace",
24 "description": "Complete problems 1-5.",
25 "weight_percentage": 5,
26
27 "next_steps": [
28   { "text": "Read Chapter 4", "is_completed": true }
29 ]
30 }
```